

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0706

COURSE OUTCOME COVERED

C02: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

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Annexure A

Debugging & Optimisation Task

Code of Conduct:

I M Nishanth (192511154) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M Nishanth

Annexure A

Debugging & Optimisation Task

1. A rapidly growing IT company has shifted its operations to a newly constructed sixstorey smart office building. More than **1,200 employees** use wireless laptops, smartphones, IP phones, printers, and IoT devices connected through the company's **Wireless Local Area Network (WLAN)**. The office relies heavily on cloud computing, online meetings, video conferencing, and file sharing, making uninterrupted wireless connectivity essential for daily operations. In addition, the company has a team of field engineers and sales executives who travel across different cities and use **4G/5G Wireless Wide Area Network (WWAN)** services to access the company's servers, customer databases, and business applications from remote locations. During the last few weeks, employees have reported several connectivity issues. Users experience slow internet speeds during peak working hours, frequent Wi-Fi disconnections while moving between floors, and weak wireless signals in conference rooms. Video conferencing applications often freeze or disconnect, affecting communication with international clients. Network administrators have also observed that some wireless access points are overloaded while others remain underutilized, leading to uneven traffic distribution. Furthermore, field engineers working in rural and highway areas frequently lose WWAN connectivity, delaying the upload of maintenance reports and customer information. Security audits reveal that a few wireless devices are still connected using outdated security protocols, exposing sensitive company information to potential cyber threats. The management has requested the network engineering team to investigate these issues, identify the root causes, and optimize both the WLAN and WWAN infrastructure to ensure high performance, secure communication, and seamless connectivity for all users.

Ans: Enterprise WLAN & WWAN Connectivity Optimization

1. Problem Identification (Root Causes)

- Poor AP placement and channel planning – causes co-channel interference between floors and dead zones in conference rooms.
- Insufficient AP density for 1,200+ devices, leading to AP overload on some access points while others are underused.

- No load balancing / band steering between APs, so clients cluster onto a few access points during peak hours.
- Lack of fast-roaming support (802.11k/v/r), causing disconnections as employees move between floors.
- Legacy security protocols (WEP / WPA-Personal) still active on some devices, exposing the network to attacks.
- WWAN dropouts for field staff are due to weak carrier signal in rural/highway zones and no failover between carriers.

2. Debugging Approach

- Conduct a full RF site survey / heat-map analysis per floor to visualize signal strength, interference, and channel overlap.
- Pull AP utilization logs from the Wireless LAN Controller (WLC) to identify overloaded vs underutilized access points.
- Audit security configuration on all APs and endpoints to flag devices still using outdated encryption (WEP/WPA-TKIP).
- Review WWAN signal logs and carrier coverage maps for field engineer routes to identify dead zones.
- Use packet capture on video-conferencing sessions to confirm packet loss/jitter correlates with Wi-Fi roaming events.

3. Optimization Strategies

- Redesign AP placement using survey data; add APs in conference rooms and stairwells to eliminate dead zones.
- Enable band steering and client load balancing on the WLC so clients are distributed evenly between 2.4 GHz and 5/6 GHz radios.
- Enable 802.11k/v/r for fast, seamless roaming between floors without dropped calls.
- Apply Quality of Service (WMM/DSCP marking) to prioritize video conferencing and VoIP traffic over bulk downloads.
- Upgrade all wireless security to WPA3-Enterprise with 802.1X authentication; disable legacy WEP/WPA-TKIP support entirely.

- For WWAN: deploy dual-SIM/multi-carrier failover routers and SD-WAN with cellular bonding for field engineers; enable offline data caching that syncs automatically once connectivity is restored.

4. Sample Configuration (Illustrative WLC / Switch Commands)

```
! -- Enable band steering and load balancing on Wireless LAN
Controller --

wlan load-balancing enable

wlan band-select enable


! -- Enable fast roaming (802.11r) on the SSID --

wlan security wpa akm 802.1x-ft enable


! -- QoS: prioritize video conferencing traffic --

class-map match-any VIDEO_CONF
    match dscp af41
policy-map WLAN_QOS
    class VIDEO_CONF
        priority percent 30


! -- Enforce WPA3-Enterprise, disable legacy security --

wlan security wpa3 enable

wlan security wpa wpa2 disable
```

5. Analysis & Justification

Redesigning AP placement and enabling load balancing directly resolves uneven traffic distribution and dead zones. Fast roaming (802.11k/v/r) eliminates the disconnect-reconnect delay employees experience moving between floors, which is what causes video calls to freeze. QoS prioritization ensures conferencing traffic is not starved during peak congestion. Upgrading to WPA3-Enterprise closes the security gap identified in the audit. Finally, multi-carrier WWAN failover with offline caching ensures field engineers can keep working and sync data automatically even in poor-coverage rural areas, addressing the last root cause without any single point of failure.

2. A multinational financial organization operates its headquarters, regional offices, and multiple data centers using an **Asynchronous Transfer Mode (ATM) network** to support mission-critical services such as online banking, video conferencing, Voice over IP (VoIP), and real-time financial transactions. The ATM network uses fixed-size **53-byte cells** and establishes **Virtual Circuits (VCs)** to provide reliable and predictable Quality of Service (QoS). Recently, the organization has experienced several performance issues during peak business hours. Customers report delays in online transactions, voice calls suffer from jitter and occasional packet loss, and video conferencing sessions experience reduced quality. Network monitoring indicates that several ATM switches are operating at high utilization, some Virtual Circuits are congested, and available bandwidth is not being efficiently allocated among different traffic classes. Engineers also discover that certain applications are using inappropriate ATM service categories, leading to increased delay and inefficient resource utilization. The IT department has assigned you as the network engineer to investigate the ATM network, identify the faults, debug the performance issues, and recommend optimization strategies to restore efficient, reliable, and high-quality communication across the enterprise.

Ans: ATM Network Performance Debugging

1. Problem Identification (Root Causes)

- Applications are mapped to incorrect ATM service categories – e.g., VoIP or video may be running over UBR (Unspecified Bit Rate) instead of CBR/rt-VBR, causing jitter and cell loss.
- Several ATM switches are near maximum utilization, causing queuing delay and cell loss during peak hours.
- VCs are congested because Peak Cell Rate (PCR) / Sustainable Cell Rate (SCR) parameters are not properly provisioned or enforced.
- Bandwidth is not efficiently partitioned among traffic classes, so bursty best-effort traffic can starve real-time traffic.

2. Debugging Approach

- Collect per-VC statistics: Cell Loss Ratio (CLR), Cell Delay Variation (CDV), and Cell Transfer Delay (CTD) from switch monitoring tools.
- Identify which VCs carry VoIP/video and verify their assigned ATM service category (CBR, rt-VBR, nrt-VBR, ABR, UBR).

- Check traffic shaping/policing configuration (PCR, SCR, Maximum Burst Size – MBS) at the User-Network Interface (UNI) against actual traffic profiles.
- Review switch utilization reports to find congestion hotspots and links operating above safe thresholds (commonly > 80%).

3. Optimization Strategies

- Re-map traffic to correct ATM service categories: CBR for VoIP (guaranteed bandwidth, minimal jitter), rt-VBR for video conferencing, nrt-VBR/ABR for banking transactions, UBR only for best-effort/background transfers.
- Configure accurate PCR/SCR/MBS values per VC to match real traffic demand and enable proper traffic shaping/policing at the UNI.
- Implement Connection Admission Control (CAC) to prevent over-provisioning of VCs beyond available switch capacity.
- Load-balance VCs across multiple physical trunks/switches, or upgrade trunk bandwidth on the most congested links.
- Continuously monitor QoS metrics (CLR, CDV, CTD) and dynamically rebalance VC allocations as usage patterns shift.

4. Sample Configuration (Illustrative ATM VC Provisioning)

```
! -- Configure VoIP over a CBR virtual circuit --
interface ATM0/0
  pvc 0/101
    vbr-nrt 0                ! remove any incorrect VBR mapping
    cbr 1536                 ! Constant Bit Rate: 1536 kbps guaranteed for
VoIP
    broadcast

! -- Configure video conferencing over rt-VBR --
pvc 0/102
  vbr-rt 2000 1500 32      ! PCR=2000kbps, SCR=1500kbps, MBS=32 cells

! -- Banking transaction traffic over nrt-VBR (reliable, non-real-
time) --
pvc 0/103
```

5. Analysis & Justification

Matching each traffic type to its correct ATM service category is the single biggest fix here: CBR guarantees fixed bandwidth for VoIP so jitter disappears, and rt-VBR gives video conferencing bounded delay while still allowing some burst tolerance. Correct PCR/SCR/MBS values combined with Connection Admission Control prevent the network from accepting more traffic than switches can handle, directly reducing the high utilization and VC congestion observed. Load balancing across trunks spreads the peak-hour load so no single switch becomes a bottleneck, restoring reliable, low-jitter performance for transactions, voice, and video.

3. A multinational e-commerce company has interconnected its headquarters, regional offices, warehouses, and cloud data centers using a **Multiprotocol Label Switching (MPLS)** network to ensure fast, secure, and reliable communication. The MPLS infrastructure supports critical business applications, including online order processing, enterprise resource planning (ERP), video conferencing, Voice over IP (VoIP), cloud-based inventory management, and real-time customer support. Recently, the organization has experienced frequent performance issues during peak business hours. Employees report slow access to cloud applications, delayed order processing, poor-quality voice calls, and interruptions during video conferences. Network monitoring reveals that some Label Switched Paths (LSPs) are heavily congested, while other paths remain underutilized. The IT team also discovers that several routers have incorrect label mappings, causing inefficient routing and increased latency. In addition, traffic engineering has not been properly configured, resulting in uneven bandwidth utilization across the MPLS backbone. Since these issues are affecting customer satisfaction and business operations, the network administrator has asked you to investigate the MPLS network, identify the faults, debug the performance problems, and recommend suitable optimization techniques to improve the overall efficiency and reliability of the network.

Ans: MPLS Backbone Fault Debugging & Traffic Engineering

1. Problem Identification (Root Causes)

- Incorrect label mappings on several routers cause packets to be forwarded along inefficient or looping paths, increasing latency.

- Absence of MPLS Traffic Engineering (TE) means all traffic follows the IGP shortest path, overloading some LSPs while others sit idle.
- No bandwidth reservation or Constraint-Based Routing (CSPF) is configured, so critical applications (VoIP/video) compete unfairly with bulk data traffic.
- Lack of monitoring means congestion hotspots are not detected and corrected proactively.

2. Debugging Approach

- Verify LDP/RSVP-TE label bindings at each router using label-forwarding table checks to detect mismatches.
- Trace actual packet paths with LSP ping/traceroute to confirm whether traffic follows the intended path or is misrouted.
- Analyze per-LSP bandwidth utilization (via SNMP/NetFlow) to identify which paths are congested and which are underused.
- Audit router configuration for missing traffic-engineering extensions (RSVP-TE, CSPF) and QoS/EXP-bit marking.

3. Optimization Strategies

- Correct the faulty label bindings identified during debugging so packets follow their intended LSPs without misrouting.
- Enable MPLS Traffic Engineering (RSVP-TE) with explicit/constraint-based paths so traffic is distributed across all backbone links rather than only the IGP shortest path.
- Reserve bandwidth per LSP for critical applications and use MPLS QoS (EXP-bit / DiffServ-aware TE) to prioritize VoIP and video over bulk ERP/data traffic.
- Enable load balancing across multiple LSPs (or ECMP) between the same source-destination pair to use underutilized paths.
- Deploy MPLS Fast Reroute (FRR) for automatic failover, and continuously monitor LSP utilization to rebalance traffic proactively.

4. Sample Configuration (Illustrative MPLS TE Setup)

```
! -- Enable MPLS traffic engineering globally and per interface --
mpls traffic-eng tunnels
interface GigabitEthernet0/0
  mpls traffic-eng tunnels
```



```

ip rsvp bandwidth 1000000

! -- Define an explicit low-latency path for VoIP/video traffic --
ip explicit-path name VOICE_VIDEO_PATH enable
  next-address 10.1.1.2
  next-address 10.1.2.2

! -- Create a TE tunnel with bandwidth reservation and priority --
interface Tunnel0
  tunnel mode mpls traffic-eng
  tunnel mpls traffic-eng bandwidth 500000
  tunnel mpls traffic-eng path-option 1 explicit name
VOICE_VIDEO_PATH
  tunnel mpls traffic-eng priority 1 1

```

5. Analysis & Justification

Fixing the incorrect label mappings removes the misrouting that was inflating latency. Introducing RSVP-TE with explicit paths and bandwidth reservation directly resolves the imbalance between congested and idle LSPs by actively steering traffic instead of relying purely on IGP shortest-path routing. Reserving bandwidth and prioritizing EXP bits for VoIP/video ensures these latency-sensitive applications are protected even during peak load, while Fast Reroute adds resilience against link failures. Together these measures restore even utilization across the MPLS backbone and improve the customer-facing application performance.

4. A multinational logistics company operates an automated warehouse where barcode scanners, RFID readers, autonomous robots, and inventory management systems continuously exchange data over a computer network. Every day, thousands of shipment records, customer details, and inventory updates are transmitted between warehouse terminals and the central server. Recently, the company has observed that some shipment records are arriving with incorrect information, barcode scans occasionally fail, and inventory quantities in the database do not always match the physical stock. During peak hours, network monitoring reports an increase in transmission errors caused by electrical interference, signal attenuation, and network congestion. Although some corrupted frames are detected and retransmitted, others remain undetected, resulting in incorrect database entries and delayed order processing. The IT department suspects that the current error detection and correction mechanisms are not

configured efficiently. As the network engineer, you are assigned to investigate the transmission errors, identify the weaknesses in the existing error control techniques, and recommend suitable debugging and optimization strategies to improve the reliability and accuracy of data communication.

Ans: Warehouse Network Error Control Debugging

1. Problem Identification (Root Causes)

- The current error-detection mechanism (likely a simple parity check or weak checksum) is not strong enough to catch multi-bit errors caused by EMI and attenuation.
- No Forward Error Correction (FEC) is in place, so every detected error requires a retransmission, and some errors go entirely undetected, corrupting the database.
- Electrical/electromagnetic interference from heavy warehouse machinery and robots is degrading the signal on wireless scanner/reader links.
- Network congestion during peak hours increases collisions and buffer drops, compounding the error rate.

2. Debugging Approach

- Analyze frame error rate and error logs per device/segment to isolate which readers/robots see the highest error rates.
- Correlate error spikes with known interference sources (motor start-up times, robot movement near scanners).
- Test the current error-detection scheme by deliberately injecting known bit errors and checking whether they are always flagged.
- Measure congestion levels (collision rate, buffer occupancy) during peak periods to quantify the congestion-related contribution to errors.

3. Optimization Strategies

- Replace the weak checksum/parity scheme with a strong Cyclic Redundancy Check (CRC-32), which reliably detects burst errors common in noisy industrial environments.

- Introduce Forward Error Correction (e.g., Hamming code) on critical short-frame traffic (barcode/RFID data) so single-bit errors are corrected instantly without waiting for retransmission.
- Implement Automatic Repeat reQuest (Selective-Repeat ARQ) so only the specific corrupted frames are retransmitted, rather than entire blocks, reducing redundant traffic.
- Address the physical layer causes: use shielded cabling, reposition wireless readers away from motors/robots, and increase transmit power or reduce cell size for wireless coverage in high-interference zones.
- Segment the network (VLANs) to separate scanner/reader traffic from robot control and server traffic, reducing congestion-induced errors.

4. Sample Implementation (Error Detection / Correction Logic)

```
// -- Frame integrity check using CRC-32 before accepting warehouse
data --

function isFrameValid(frameBits, receivedCRC) {
  // Compute CRC-32 over received data bits
  const computedCRC = calculateCRC32(frameBits);
  return computedCRC === receivedCRC;    // true = frame accepted
}

// -- Hamming(12,8) single-bit correction for short barcode/RFID
frames --

function correctSingleBitError(codeword) {
  const syndrome = computeParitySyndrome(codeword); // c8 c4 c2 c1
  if (syndrome !== 0) {
    codeword[syndrome - 1] ^= 1;    // flip the faulty bit position
  }
  return codeword;
}
```

5. Analysis & Justification

Upgrading from a weak checksum to CRC-32 closes the gap where corrupted frames were previously going undetected, since CRC-32 reliably catches burst errors typical of EMI. Adding Hamming-code FEC for short, latency-sensitive scans allows single-bit errors to be corrected

instantly, cutting down on retransmission delay for barcode/RFID reads. Selective-Repeat ARQ ensures only genuinely bad frames are resent, reducing unnecessary network load. Finally, addressing the physical root causes (shielding, repositioning, VLAN segmentation) reduces the error rate at the source rather than just handling it after the fact – together restoring accurate inventory and shipment data.

5. A leading online education platform conducts live classes and online examinations for more than **15,000 students** across different locations. During examinations, the central server simultaneously sends question papers, images, videos, and answer submission acknowledgements to all students over the network. The server is equipped with a highspeed **10 Gbps** connection, whereas many students access the system using slower broadband or mobile internet connections. During peak hours, students experience incomplete downloads, delayed page loading, duplicate packets, and repeated retransmissions. Some students receive the next set of data before their devices finish processing the previous one, causing receiver buffer overflow and application crashes. Network administrators also observe an excessive number of retransmissions, increased latency, reduced throughput, and poor bandwidth utilization. The current flow control mechanism is not effectively regulating the transmission rate between the sender and receivers. As the network engineer, you have been assigned to investigate the communication problems, identify the causes of inefficient flow control, debug the existing system, and recommend suitable optimization techniques to ensure reliable and efficient data transmission during high-traffic periods.

Ans: Online Examination Platform Flow-Control Debugging

1. Problem Identification (Root Causes)

- Severe speed mismatch between the 10 Gbps server and slower student broadband/mobile connections is not being compensated for by flow control.
- The server is not correctly honoring the receiver's advertised window size (rwnd), so it keeps sending data faster than clients can consume it.
- Congestion control on the server may not be adapting its sending rate based on real-time network/receiver conditions (e.g., stuck in an aggressive send mode).
- A single centralized server pushing content directly to 15,000 clients creates a scalability bottleneck with no load distribution.

2. Debugging Approach

- Capture and analyze TCP handshake and window-size fields during a live session to confirm whether the server is respecting the receiver's advertised window.
- Check for excessive duplicate ACKs and retransmission timeouts, which indicate the congestion window is not adapting properly.
- Profile receiver-side buffer occupancy on slow-connection test devices to pinpoint the exact overflow threshold.
- Review server-side socket/buffer and congestion-control algorithm configuration (e.g., default vs. tuned settings).

3. Optimization Strategies

- Ensure strict compliance with TCP flow control: the server must throttle transmission based on each client's advertised receive window (rwnd), not just its own capacity.
- Enable TCP window scaling and use an adaptive congestion-control algorithm (e.g., BBR) so the sending rate matches actual path and receiver capacity rather than always pushing at maximum speed.
- Introduce server-side per-client rate limiting/traffic shaping so no single high-bandwidth burst overwhelms a slow client.
- Distribute load using a Content Delivery Network (CDN) or edge caching/regional mirror servers, instead of one central server serving all 15,000 students directly.
- Use adaptive content delivery (e.g., compressed/lower-resolution images for slow connections) and exponential backoff for retransmissions instead of immediate aggressive resends.

4. Sample Implementation (Adaptive Rate Control Logic)

```
// -- Throttle server send rate based on each client's advertised
window --

function sendData(client, dataQueue) {
    const sendable = Math.min(client.advertisedWindow,
dataQueue.length);

    transmit(client, dataQueue.slice(0, sendable));    // respect rwnd,
never exceed it

    dataQueue = dataQueue.slice(sendable);
}
```

```
// -- Exponential backoff for retransmissions instead of immediate
resend --

function retransmitWithBackoff(packet, attempt) {
  const delay = BASE_DELAY_MS * Math.pow(2, attempt); // 100ms,
200ms, 400ms...
  setTimeout(() => transmit(packet), delay);
}
```

5. Analysis & Justification

Enforcing strict flow control – where the server never sends more than a client's advertised window can hold – directly eliminates receiver buffer overflow and the resulting application crashes. Adaptive congestion control (e.g., BBR) lets the sending rate self-adjust to real path/receiver conditions instead of assuming every client can handle 10 Gbps. Distributing delivery through a CDN or regional mirrors removes the single-server bottleneck, so 15,000 simultaneous students are served from multiple closer points rather than one origin. Exponential backoff on retransmissions further reduces duplicate-packet storms during congestion. Together, these changes restore reliable, efficient delivery of exam content regardless of each student's connection speed.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, wellstructured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation